

Eliezio Oliveira

Lead Software Engineer

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PROFESSIONAL SUMMARY

Lead software engineer with 20+ years building distributed backend systems in Java, Kotlin, and Spring Boot. Deep experience in microservices, REST and GraphQL API design, PostgreSQL performance tuning, applied cryptography and HSM integration, and AWS/Kubernetes delivery. Author of two open-source libraries on Maven Central: dbQueryWatch, which catches slow queries before they reach production, and SLog4j, for structured event logging. Strong computer science foundation, broad technical range, and a relentless focus on designing and delivering robust, well-crafted solutions.

EXPERIENCE

Senior Software Engineer 🗓 Jan 2026 - Present

[Fidelity Investments](#) | A global investment and financial services firm | Galway, Ireland (Hybrid)

- Cut database round-trips on the heaviest configuration endpoints from up to 3,300 queries per request to 1, replacing lazy-loading cascades with JPA entity graphs, and built a compatibility test kit (TCK) that verified the optimization left the REST API and Kafka messaging contracts untouched, comparing every response and published payload against the pre-change build.
- Diagnosed production incidents and cross-environment performance discrepancies across a Java and Spring microservice estate, investigating container restarts and using Glowroot to trace query-time differences between pre-production and production.



Skills: Java 17, Groovy, Shell Script, Spring Boot, Microservices, JPA/Hibernate, Query Optimization, Glowroot, Oracle, Kafka, REST API, API Gateway, Angular, TypeScript, Cypress, WireMock, Spock Framework, TDD, ATDD.

Technical Consultant 🗓 Apr 2025 - Sep 2025

[RioCard Tecnologia](#) | Electronic ticketing tech for public transport | Rio de Janeiro, Brazil (Remote)

- Led the migration of cryptographic services from IBM z/OS ICSF to cloud HSMs (AWS CloudHSM, later Oracle OCI Vault), enabling client applications to start migration three months earlier through HSM abstraction and simplified integration.
- Designed and built CryptoApi, the Quarkus service behind that migration: an API-first REST surface covering card security, digital signature, key distribution and receipt operations, authenticated by mutual TLS with per-client certificates and per-client endpoint authorization across the five consuming applications.
- Shipped a Java 8-compatible client SDK generated from the OpenAPI specification, plus an Oracle PL/SQL integration path backed by Oracle Wallet, so both legacy applications and database stored procedures could adopt the service without changing runtime or platform.
- Enforced the API contract in CI with Spectral linting, Pact consumer-driven contracts and Schemathesis property-based testing in both positive and negative modes, holding quality with ArchUnit rules, Pitest mutation testing and 100% coverage on application packages.
- Automated security verification as part of the test suite, covering TLS posture and a role-versus-endpoint authorization matrix.
- Partnered with the Information Security Group to design and conduct a new key ceremony, ensuring secure key management and compliance during the transition to cloud environments.



Skills: Java 17, Quarkus, Microservices, MicroProfile Fault Tolerance, Hexagonal Architecture,DDD, REST API, API First, OpenAPI, mTLS, Oracle PL/SQL, Pact, Schemathesis, ArchUnit, Mutation Testing (Pitest), Gatling, HashiCorp Vault, AWS CloudHSM, Oracle OCI, JOSE/JWT, Docker, Kubernetes, GitLab CI, SonarQube, Error Prone, NullAway, TDD, ATDD, Resilience Testing, Prometheus, OpenTelemetry, Grafana.

Senior Backend Software Engineer 🏢 Apr 2022 - Mar 2025

[Beeline Inc.](#) | Global provider of workforce management solutions | Dublin, Ireland

- Led the campaign against N+1 query patterns across the platform, using Glowroot to trace and diagnose them in the JPA layer.
- Eliminated N+1 in the GraphQL layer by introducing data loaders, driving adoption by coaching developers and providing implementation guidance.
- Helped replace JPA with jOOQ on complex and performance-critical queries, improving query efficiency and maintainability.
- Built [dbQueryWatch](#) to shift slow-query detection left into the test suite: it analyses the execution plan of every query a test class issues and fails the build on full table scans or excessive cost. Open-sourced and published to Maven Central, supporting PostgreSQL, MySQL and Oracle.
- Took part in the production on-call rotation, diagnosing and resolving live messaging incidents across Kafka and ActiveMQ, including dead-letter queue (DLQ) backlogs, poison-pill serialization failures, and bloated queues.



Skills: Kotlin, Spring Boot, SQL, Microservices, GraphQL, REST API, PostgreSQL, JPA/Hibernate, jOOQ, Query Optimization, Glowroot, Kafka, ActiveMQ, AWS, Docker, Kubernetes, Testcontainers, CI/CD, Gradle, Gitlab, On-Call, Incident Response.

Senior Backend Engineer 🏢 Aug 2021 - Mar 2022

[Teckro Ltd](#) | Clinical trial software platform for pharma and CROs | Limerick, Ireland

Improved server–client integration reliability and eliminated contract mismatches through automation and formalized API governance.

- Introduced a CI-validated OpenAPI specification to formalize server–client contracts, eliminating recurrent integration issues.
- Partnered with front-end developers to automate the generation process using JavaScript scripts, ensuring synchronized schemas and faster delivery cycles.



Skills: Kotlin, Spring Boot, Microservices, OpenAPI 3, REST API, API First, MySQL, TypeScript, Docker

Contract Senior DevOps Engineer 🏢 Aug 2020 - Jul 2021

[Aetion Inc.](#) | US real-world evidence platform for biopharma and payers | New York, USA (Remote)

- Designed and built [SARI](#) (Secure Access to RDS Instances), a declarative access-management system that removed standing database credentials for analysts: permissions, grant levels and validity windows live as version-controlled YAML, and the system reconciles them across MySQL grants, IAM, AWS Glue connections and bastion-host SSH keys.
- Sourced identity from Okta and scheduled each reconciliation run from the next permission transition via CloudWatch, so access expires and renews with no manual intervention and no credential outlives its window.
- Built a self-service web application for analysts to raise access requests, replacing an ad-hoc process while keeping every change reviewable and auditable in Git.



Skills: Python 3, Pulumi, Terraform, MySQL, SAML, AWS (IAM, Lambda, RDS, Glue, CodeBuild, ECR, S3, CloudWatch), Infrastructure-as-Code, GitOps, Okta, SSH, Docker, Testcontainers, CI/CD, pytest, Flask, JavaScript, jQuery.

Senior Engineer 🏢 Jan 2019 - Jul 2020

Contributed upstream to ONAP Controller Design Studio (CDS), an open-source platform for telecom network automation.

- Designed and built the [User Acceptance Test \(UAT\) framework](#) [🔗](#) for the CDS Blueprint Processor: declarative, data-driven specifications in YAML that run the service in a near-production configuration against an in-memory database with every external service mocked. Still maintained upstream today and now covering four blueprint models.
- Added a /spy endpoint that records a blueprint run against live external services and emits a ready-to-use test specification, removing the need to hand-write them.
- Rehabilitated the upstream test suite, re-enabling long-disabled tests across the NETCONF, gRPC and REST client modules, fixing order-dependent failures, and clearing bugs and vulnerabilities reported by SonarQube.
- Integrated the UATs into the CI/CD pipeline, raising release confidence and cutting debugging cycles across frequent upstream releases. Contributions: github.com/onap/ccsdk-cds [🔗](#).



Skills: Kotlin, Spring Boot, Python 3, Maven, YAML, H2, REST API, gRPC, NETCONF, Docker, CI/CD, Jenkins, Gerrit, SonarQube, TDD, ATDD, WireMock.

Lead Software Engineer Aug 2016 - Aug 2018

[RioCard Tecnologia](#) [🔗](#) | Electronic ticketing tech for public transport | Rio de Janeiro, Brazil

Extended and hardened the online transaction platform built during my first tenure.

- Raised automated coverage of the main application to nearly 70%, using end-to-end tests with minimal mocking so the suite stayed representative of production behaviour.
- Established production observability for online card operations, instrumenting the platform and adding Prometheus and Grafana monitoring.
- Created and adopted [SLog4j](#) [🔗](#), a structured event logging library for Java that replaces prose log messages with machine-parseable key-value events. Open-sourced and published to Maven Central.
- Moved the client-server protocol to WebSockets, making communication HTTP-friendly and simplifying client integration.



Skills: Java, Spring Boot, REST API, WebSockets, SLF4J, Structured Logging, Spock Framework, TDD, End-to-End Testing, Micrometer, Prometheus, Grafana, Jenkins.

Lead Software Engineer Aug 2014 - Aug 2016

[Muxi Tecnologia em Pagamentos S.A.](#) [🔗](#) | ^[1] | Rio de Janeiro, Brazil

- Tech-led a 6-engineer team designing and building a payment scheme to strengthen the security of web-based credit card transactions, owning the architecture from protocol design through implementation.



Skills: Java, Spring, Angular, REST API, Payment Security, Cryptography, POS Terminals.

Senior Software Engineer Aug 2008 - Aug 2014

[RioCard Tecnologia](#) [🔗](#) | Electronic ticketing tech for public transport | Rio de Janeiro, Brazil

- Re-architected online contactless card operations, moving database access and Mifare key diversification to the server and leaving only a hardened reader in the field, removing two attack surfaces: database connections exposed in open environments and theft-prone Security Access Modules (SAMs).
- Cut online top-up from up to 60 seconds to roughly 1 second at peak load by deferring updates to high-contention database tables into batch operations, sustaining near-100% reliability.
- Extended the platform to online debit and large-scale card issuance, and integrated third-party ATM providers with minimal exposure of card-handling operations, reaching hundreds of third-party and public venues.
- Designed a Lua-based middleware for POS terminals deployed to hundreds of devices across Metro stations, speeding delivery of ticketing applications and enabling automatic remote updates that removed on-site maintenance across the Rio de Janeiro metropolitan area.
- Tech-led a 4-engineer team building the web applications that operated in tandem with the online transaction platform.



Skills: Java, Netty, Java Cryptography Extension, TLS, HSM, SAM, Mifare Classic, MyBatis, Lua, C, Google C++ Test Framework, SOLID, JavaScript, jQuery.

Senior Software Engineer 🏢 Aug 2004 - Aug 2008

[Muxi Tecnologia em Pagamentos S.A.](#) [🔗](#) (formerly APPI Tecnologia) | ^[1] | Rio de Janeiro, Brazil

- Architected and led the development of the first wide-deployed communication stack (TCP/IP, TLS/mTLS) for the [POSWEB™](#) [🔗](#) platform, deployed to more than two million devices in Brazil and other countries.
- Architected and developed a high-performance and high-available (99.999% uptime) middleware to integrate more than 1 million mobile POS terminals.



Skills: C, TCP/IP, TLS, Cryptography.

EDUCATION

Bachelor of Computer Science 🏢 Apr 2006

[Federal University of Rio de Janeiro, Institute of Computing](#) | Ranked 5th in QS Latin America 2025 [🔗](#)



Final Year Project: [Implementing TLS Extensions on OpenSSL \(RFC 3546\)](#) [🔗](#)

CERTIFICATIONS

- [Certified Developer for Apache Kafka \(Confluent\)](#)
- [Parallel, Concurrent, and Distributed Programming in Java \(Rice University\)](#)
- [Cryptography I \(Stanford\)](#)

LANGUAGES

- English: Full Professional Proficiency (C1 Level)
- Portuguese: native